

Deep Research

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Introduction

- LLMs can store a lot of knowledge but can struggle to ground their answers in factual, up-to-date information.
- “Deep Research” = systems that can search, reason, and self-correct across multiple sources.
- Modern AI assistants (ChatGPT Search, Gemini Deep Research, Bing Copilot) all rely on this idea.

Today's Papers

- RAG (2020): LMs + retrieval.
- Self-RAG (2023): Reflect and critique retrieval.
- MADAM-RAG (2025): Multi-agent reasoning to address problems.
- GAIA (2024): Benchmark for real-world AI Systems.

Retrieval-Augmented Generation (RAG) for Knowledge-Intensive NLP Tasks

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis,
Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, Douwe Kiela

LLMs

- Substantial amount of in-depth knowledge from data without access to an external memory
 - Store factual knowledge in their parameters
- Achieve SotA when fine-tuned on downstream NLP tasks
- But these models
 - Can't expand/revise their memory once trained
 - Can't provide insight into predictions or ground themselves in evidence
- Lagging performance in knowledge intensive tasks

Retrieval-Augmented Generation

- General-purpose model that combines parametric and non-parametric memory
 - Pre-trained seq2seq model
 - Vector index of Wikipedia, accessed with a pre-trained neural retriever

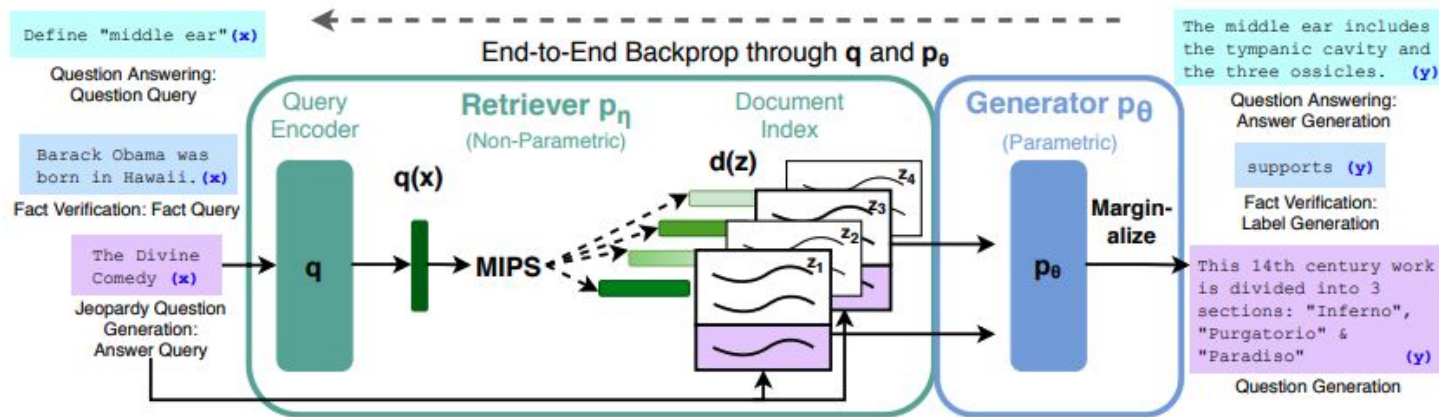


Figure 1: Overview of our approach. We combine a pre-trained retriever (*Query Encoder* + *Document Index*) with a pre-trained seq2seq model (*Generator*) and fine-tune end-to-end. For query x , we use Maximum Inner Product Search (MIPS) to find the top-K documents z_i . For final prediction y , we treat z as a latent variable and marginalize over seq2seq predictions given different documents.

Models

- 2 Approaches:
 - RAG-sequence - the model uses the same document to predict each token.
 - RAG-token - the model can use a different document for each token.
- Retriever: DPR (Dense Package Retrieval)
- Generator: BART

Experiments

- Open-domain question answering (QA)
 - Datasets: Natural Questions (NQ), TriviaQA (TQA), WebQuestions (WQ) and CuratedTrec (CT)
 - Compared against extractive methods AND “Closed-Book” approaches
- Abstractive Question Answering
 - MSMARCO
 - Task consists of question, 10 gold passages, full sentence answer annotated from passages
 - Do not supply the passages, only Q & A, to treat MSMARCO as an open-domain abstractive QA task.
 - Some questions cannot be answered without the passages, or with Wikipedia alone. Can rely on parametric knowledge here.
- Jeopardy
 - Instead of regular QA, tests open-domain question generation
 - Q-BLEU-1 metric + human evaluation
- Fact Verification
 - FEVER dataset
 - Involves classifying whether a claim is supported or refuted by Wikipedia or if there's not enough evidence

Results

Table 1: Open-Domain QA Test Scores. For TQA, left column uses the standard test set for Open-Domain QA, right column uses the TQA-Wiki test set. See Appendix D for further details.

Model		NQ	TQA	WQ	CT
Closed Book	T5-11B [52]	34.5	- /50.1	37.4	-
	T5-11B+SSM[52]	36.6	- /60.5	44.7	-
Open Book	REALM [20]	40.4	- / -	40.7	46.8
	DPR [26]	41.5	57.9 / -	41.1	50.6
RAG-Token		44.1	55.2/66.1	45.5	50.0
RAG-Seq.		44.5	56.8/ 68.0	45.2	52.2

Table 2: Generation and classification Test Scores. MS-MARCO SotA is [4], FEVER-3 is [68] and FEVER-2 is [57] *Uses gold context/evidence. Best model without gold access underlined.

Model	Jeopardy		MSMARCO		FVR3	FVR2
	B-1	QB-1	R-L	B-1	Label	Acc.
SotA	-	-	49.8*	49.9*	76.8	92.2*
BART	15.1	19.7	38.2	41.6	64.0	81.1
RAG-Tok.	17.3	22.2	40.1	41.5	72.5	<u>89.5</u>
RAG-Seq.	14.7	21.4	<u>40.8</u>	<u>44.2</u>		

Ablations

Table 6: Ablations on the dev set. As FEVER is a classification task, both RAG models are equivalent.

Model	NQ	TQA Exact Match	WQ	CT	Jeopardy-QGen		MSMarco		FVR-3 Label	FVR-2 Accuracy
					B-1	QB-1	R-L	B-1		
RAG-Token-BM25	29.7	41.5	32.1	33.1	17.5	22.3	55.5	48.4	75.1	91.6
RAG-Sequence-BM25	31.8	44.1	36.6	33.8	11.1	19.5	56.5	46.9		
RAG-Token-Frozen	37.8	50.1	37.1	51.1	16.7	21.7	55.9	49.4	72.9	89.4
RAG-Sequence-Frozen	41.2	52.1	41.8	52.6	11.8	19.6	56.7	47.3		
RAG-Token	43.5	54.8	46.5	51.9	17.9	22.6	56.2	49.4	74.5	90.6
RAG-Sequence	44.0	55.8	44.9	53.4	15.3	21.5	57.2	47.5		

Index hot-swapping

- Explores the advantage of "updatable knowledge"
- Swap old (2016) vs. new (2018) Wikipedia indices
- 70% / 68% accuracy on matching years, 4–12% on mismatched
- Updating the index = updating world knowledge (no retraining needed)

Self-RAG

LEARNING TO RETRIEVE, GENERATE, AND CRITIQUE
THROUGH SELF-REFLECTION

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, Hannaneh Hajishirzi

Problems with RAG

- Traditional RAG models always retrieve documents
 - Even when they're unnecessary or noisy.
- That causes:
 - Useless or misleading evidence
 - Over-dependence on retrieval
 - Hallucinated or unsupported claims

Self-Reflective Retrieval-Augmented Generation

- A framework that trains a single LM that adaptively retrieves passages on-demand (only when deemed necessary)
- Learns to reflect on its own generation process
 - using “**reflection**” tokens
- Allows tailoring of behavior to diverse requirements during inference

Retrieval-Augmented Generation (RAG)

Prompt How did US states get their names?

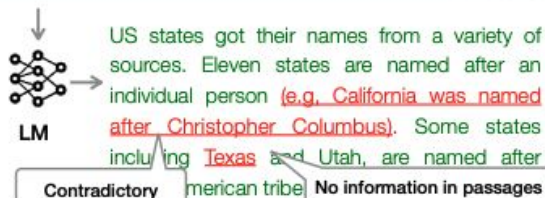
Step 1: Retrieve K documents

- 1 Of the fifty states, eleven are named after an individual person.
- 2 Popular names by states. In Texas, Emma is a popular baby name.
- 3 California was named after a fictional island in a Spanish book.

Retriever

Step 2: Prompt LM with K docs and generate

Prompt How did US states get their names? + 1 2 3



Prompt: Write an essay of your best summer vacation



Ours: Self-reflective Retrieval-Augmented Generation (Self-RAG)

Prompt How did US states get their names?

Step 1: Retrieve on demand



Step 2: Generate segment in parallel



Step 3: Critique outputs and select best segment



Prompt: Write an essay of your best summer vacation



Figure 1: Overview of SELF-RAG. SELF-RAG learns to retrieve, critique, and generate text passages to enhance overall generation quality, factuality, and verifiability.

Reflection tokens

Type	Input	Output	Definitions
Retrieve	$x / x, y$	{yes, no, continue}	Decides when to retrieve with $\mathcal{R}$
ISREL	x, d	{ relevant , irrelevant}	d provides useful information to solve x .
ISSUP	x, d, y	{ fully supported , partially supported, no support}	All of the verification-worthy statement in y is supported by d .
ISUSE	x, y	{ 5 , 4, 3, 2, 1}	y is a useful response to x .

Table 1: Four types of reflection tokens used in SELF-RAG. Each type uses several tokens to represent its output values. The bottom three rows are three types of Critique tokens, and the bold text indicates the most desirable critique tokens. x, y, d indicate input, output, and a relevant passage, respectively.

Inference Algorithm

Algorithm 1 SELF-RAG Inference

Require: Generator LM $\mathcal{M}$, Retriever $\mathcal{R}$, Large-scale passage collections $\{d_1, \dots, d_N\}$

- 1: **Input:** input prompt x and preceding generation $y_{<t}$, **Output:** next output segment y_t
 - 2: $\mathcal{M}$ predicts **Retrieve** given $(x, y_{<t})$
 - 3: **if** **Retrieve** == Yes **then**
 - 4: Retrieve relevant text passages $\mathbf{D}$ using $\mathcal{R}$ given (x, y_{t-1}) ▷ Retrieve
 - 5: $\mathcal{M}$ predicts **ISREL** given x, d and y_t given $x, d, y_{<t}$ for each $d \in \mathbf{D}$ ▷ Generate
 - 6: $\mathcal{M}$ predicts **ISSUP** and **ISUSE** given x, y_t, d for each $d \in \mathbf{D}$ ▷ Critique
 - 7: Rank y_t based on **ISREL**, **ISSUP**, **ISUSE** ▷ Detailed in Section 3.3
 - 8: **else if** **Retrieve** == No **then**
 - 9: $\mathcal{M}_{gen}$ predicts y_t given x ▷ Generate
 - 10: $\mathcal{M}_{gen}$ predicts **ISUSE** given x, y_t ▷ Critique
-

Self-RAG Training

- Generator Model **M**
 - On a curated corpus with passages from retriever **R** and reflection tokens predicted by a critic model **C**
- Critic Model **C**
 - Train **C** to generate reflection tokens for evaluating passages and task output
 - Use this model to update the training set by inserting reflection tokens into outputs offline
 - We can then train the generator model **M** as a conventional LM objective
 - No need for a critic model at inference time

Training

- Could use GPT-4 to generate reflection tokens directly
- But for scalability and reproducibility, it is only used to create supervised data to train a Critic Model C
- Critic & retriever model used to produce the training set for Generator Model M

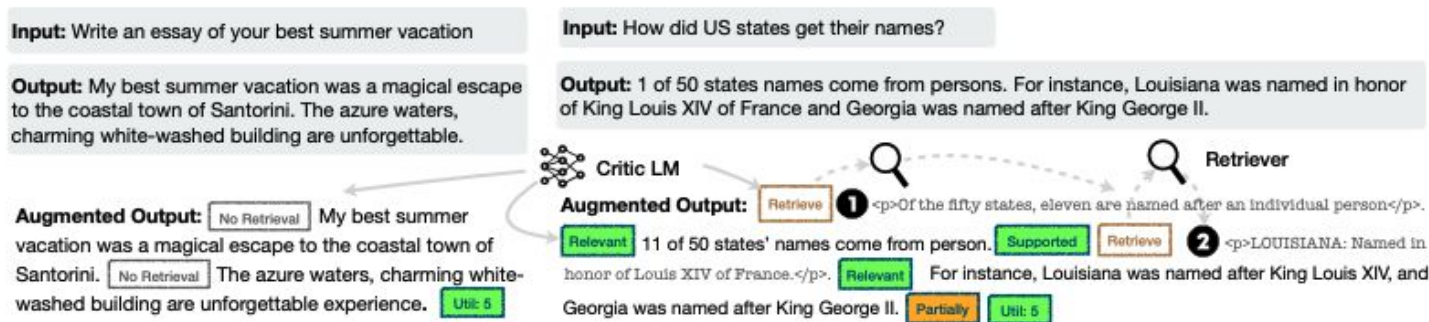


Figure 2: SELF-RAG training examples. The left example does not require retrieval while the right one requires retrieval; thus, passages are inserted. More examples are in Appendix Table 4.

Self-RAG Inference

- Generating reflection tokens to self-evaluate its own output makes it controllable during inference, tailoring to diverse task requirements:
 - For tasks demanding factual accuracy, model can retrieve passages more frequency
 - More open-ended tasks, emphasis on retrieving less and prioritizing utility score
- Approaches to enforce control to meet these distinct objectives:
 - Threshold for adaptive retrieval
 - Hyperparameters for critique tokens
 - For example: to ensure output is supported by evidence, filter out all segments where **isSup** = No Support

Experiments

- Tested on 3 types of datasets:
 - Closed-set tasks (fact verification & multiple choice reasoning)
 - Short-form generation tasks - 2 open-domain QA datasets, PopQA, TriviaQA
 - Long-form generation tasks - biography generation tasks and long-form QA tasks (ASQA)
- Compared against baselines with and without retrievals

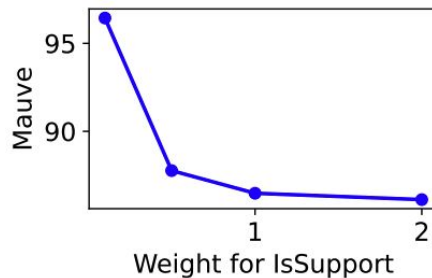
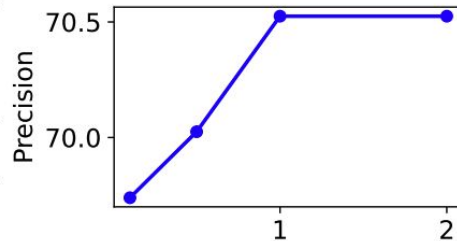
Results

LM	Short-form		Closed-set		Long-form generations (with citations)					
	PopQA (acc)	TQA (acc)	Pub (acc)	ARC (acc)	Bio (FS)	(em)	(rg)	ASQA (mau)	(pre)	(rec)
<i>LMs with proprietary data</i>										
Llama2-c _{13B}	20.0	59.3	49.4	38.4	55.9	22.4	29.6	28.6	–	–
Ret-Llama2-c _{13B}	51.8	59.8	52.1	37.9	79.9	32.8	34.8	43.8	19.8	36.1
ChatGPT	29.3	74.3	70.1	75.3	71.8	35.3	36.2	68.8	–	–
Ret-ChatGPT	50.8	65.7	54.7	75.3	–	40.7	39.9	79.7	65.1	76.6
Perplexity.ai	–	–	–	–	71.2	–	–	–	–	–
<i>Baselines without retrieval</i>										
Llama2 _{7B}	14.7	30.5	34.2	21.8	44.5	7.9	15.3	19.0	–	–
Alpaca _{7B}	23.6	54.5	49.8	45.0	45.8	18.8	29.4	61.7	–	–
Llama2 _{13B}	14.7	38.5	29.4	29.4	53.4	7.2	12.4	16.0	–	–
Alpaca _{13B}	24.4	61.3	55.5	54.9	50.2	22.9	32.0	70.6	–	–
CoVE _{65B} *	–	–	–	–	71.2	–	–	–	–	–
<i>Baselines with retrieval</i>										
Toolformer* _{6B}	–	48.8	–	–	–	–	–	–	–	–
Llama2 _{7B}	38.2	42.5	30.0	48.0	78.0	15.2	22.1	32.0	2.9	4.0
Alpaca _{7B}	46.7	64.1	40.2	48.0	76.6	30.9	33.3	57.9	5.5	7.2
Llama2-FT _{7B}	48.7	57.3	64.3	65.8	78.2	31.0	35.8	51.2	5.0	7.5
SAIL* _{7B}	–	–	69.2	48.4	–	–	–	–	–	–
Llama2 _{13B}	45.7	47.0	30.2	26.0	77.5	16.3	20.5	24.7	2.3	3.6
Alpaca _{13B}	46.1	66.9	51.1	57.6	77.7	34.8	36.7	56.6	2.0	3.8
Our SELF-RAG_{7B}	54.9	66.4	72.4	67.3	81.2	30.0	35.7	74.3	66.9	67.8
Our SELF-RAG_{13B}	55.8	69.3	74.5	73.1	80.2	31.7	37.0	71.6	70.3	71.3

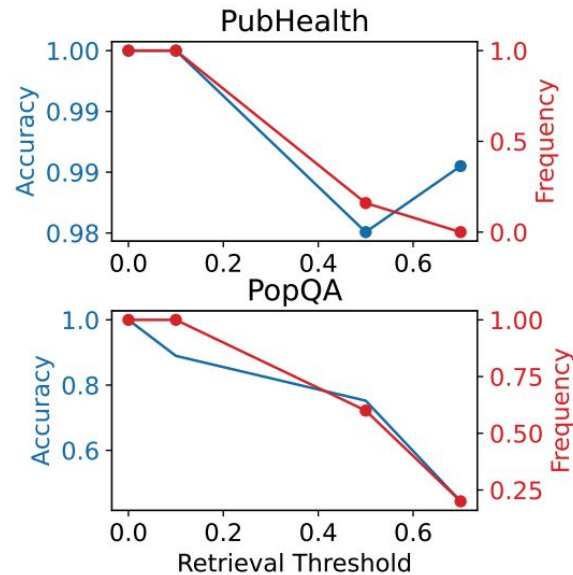
Ablations

	PQA (acc)	Med (acc)	AS (em)
SELF-RAG (50k)	45.5	73.5	32.1
<i>Training</i>			
No Retriever $\mathcal{R}$	43.6	67.8	31.0
No Critic $\mathcal{C}$	42.6	72.0	18.1
<i>Test</i>			
No retrieval	24.7	73.0	—
Hard constraints	28.3	72.6	—
Retrieve top1	41.8	73.1	28.6
Remove ISUP	44.1	73.2	30.6

(a) Ablation



(b) Customization



(c) Retrieval

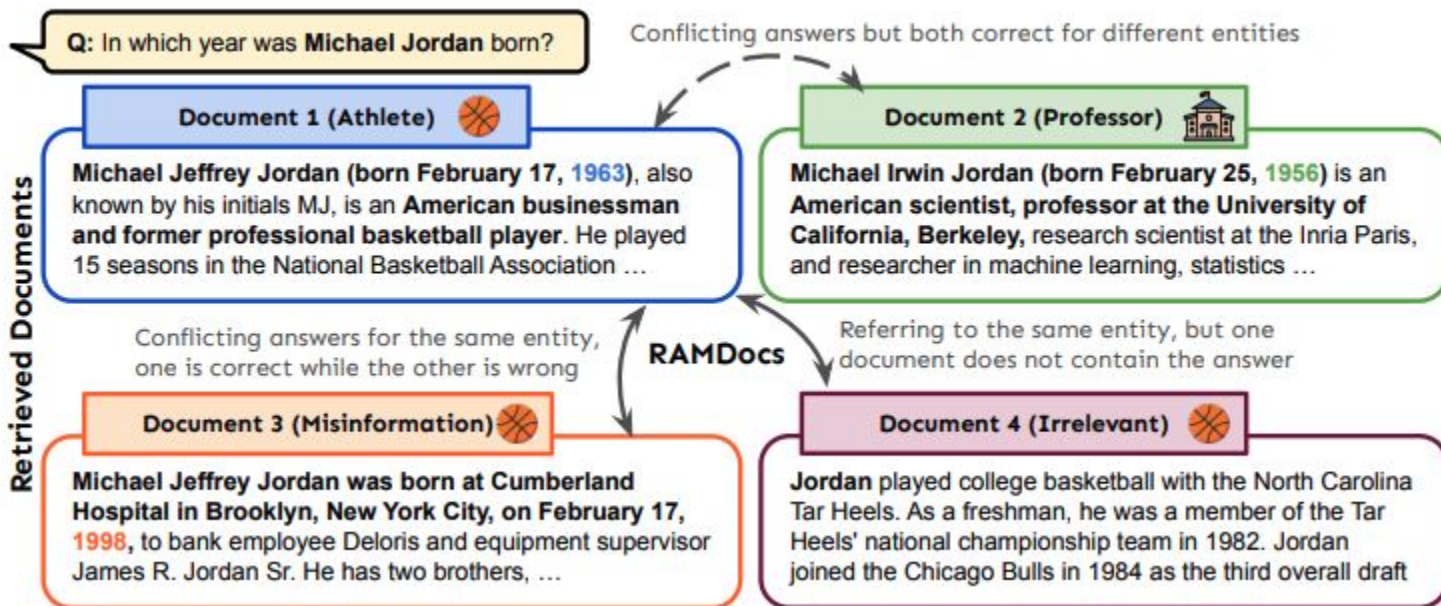
Retrieval-Augmented Generation with Conflicting Evidence

Han Wang, Archiki Prasad, Elias Stengel-Eskin, Mohit Bansal

Motivation

- LLM agents increasingly employ RAG to improve factuality, reduce hallucinations, update their parametric knowledge
- They need to be able to handle:
 - ambiguous user queries
 - conflicting information from multiple sources
 - inaccurate information from noise or irrelevant documents
- Especially when it comes to information retrieved from the internet
- Prior work usually considers these factors in isolation

In Practice



RAMDocs

Retrieval with Ambiguity and Misinformation in Documents

Dataset to simulate realistic complex scenarios for conflicting evidence, including:

- Ambiguity
- Misinformation
- Noise

Pose a challenge for existing RAG baselines by requiring handling conflicting information due to these factors simultaneously

Creating RAMDocs

- Building upon AmbigDocs, that focuses on evaluating ambiguity by presenting multiple valid answers per query
 - Total of 500 queries
 - Randomly sample 1-3 correct answers for each ambiguous query from the dataset
 - Targets the goal of ambiguous queries
 - Simulates realistic constraints of only partial information being retrieved
- Vary the number of documents supporting each valid answer for a query
 - Retrieve multiple supporting documents per query containing the correct answer
- Each query in the dataset has 1-3 answers and each answer is supported by 1-3 documents (2.20 valid answers and 3.84 valid documents per query on avg)
- 0-2 additional correct documents but with the answer replaced with something else to incorporate misinformation
- 0-2 noisy documents with low-quality or irrelevant content

MADAM-RAG

- Multi-agent framework designed to handle inter-document conflicts, misinformation, and noise in retrieved content.
- Three main components:
 - Independent LLM dialogue agents conditioned on a single document
 - Centralized aggregator
 - Iterative multi-round debate process

Single LLM Agent (1)

- Each document is assigned to an agent that processes the query with that document
- Agent generates intermediate responses based on the query and the doc independently
 - Each agent has the chance to thoroughly review the document information, reducing probability of overlooking something in long contexts
 - Prevents agent's response from being impacted by document frequency
 - In the example, "Michael I Jordan" could've been missed
- In later rounds, also receives a summary of prior rounds, for further informed decisions

Aggregator Module (2)

- Receives the outputs from all of the agents in a round
 - Agent responses are shuffled to prevent position bias
- Generates a summarized answer with explanation
- Synthesizes all the responses taking a global view over the conversation
 - Helpful to realize potential ambiguity, misinformation, and other conflicts
 - Flags 1998 as misinformation
 - For example, aggregator realizes that both 1963 and 1956 are potentially valid birth years

Multi-round Debate (3)

- In order to handle conflict, multi-round debate is conducted among the agents
- Each round, agents are given the aggregated response from the aggregator from the previous round
- At this point, they are prompted to reflect and optionally revise their own answers
- Able to defend, reflect, revise their response, helping convergence towards a consistent output

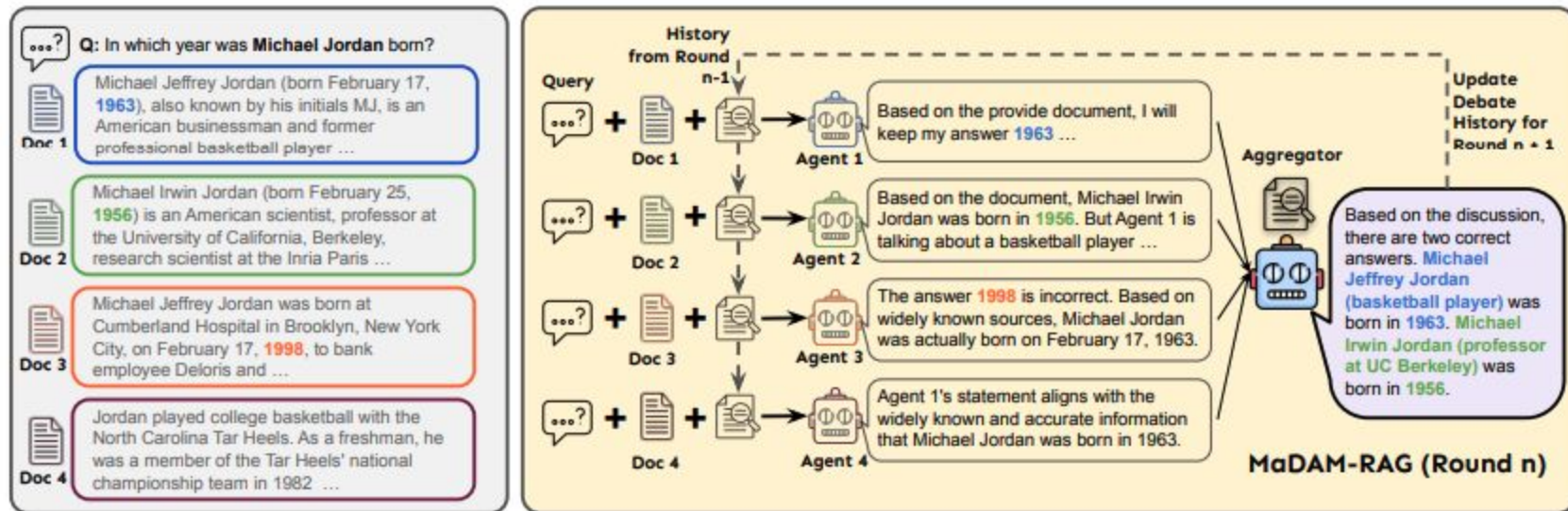


Figure 2: MADAM-RAG operates by assigning each document to a separate agent. Agents are responsible for summarizing and representing their documents, and engage in a multi-agent, multi-round debate with each other to filter out misinformation and noise and address ambiguity. An aggregator then summarizes the discussion into a single response.

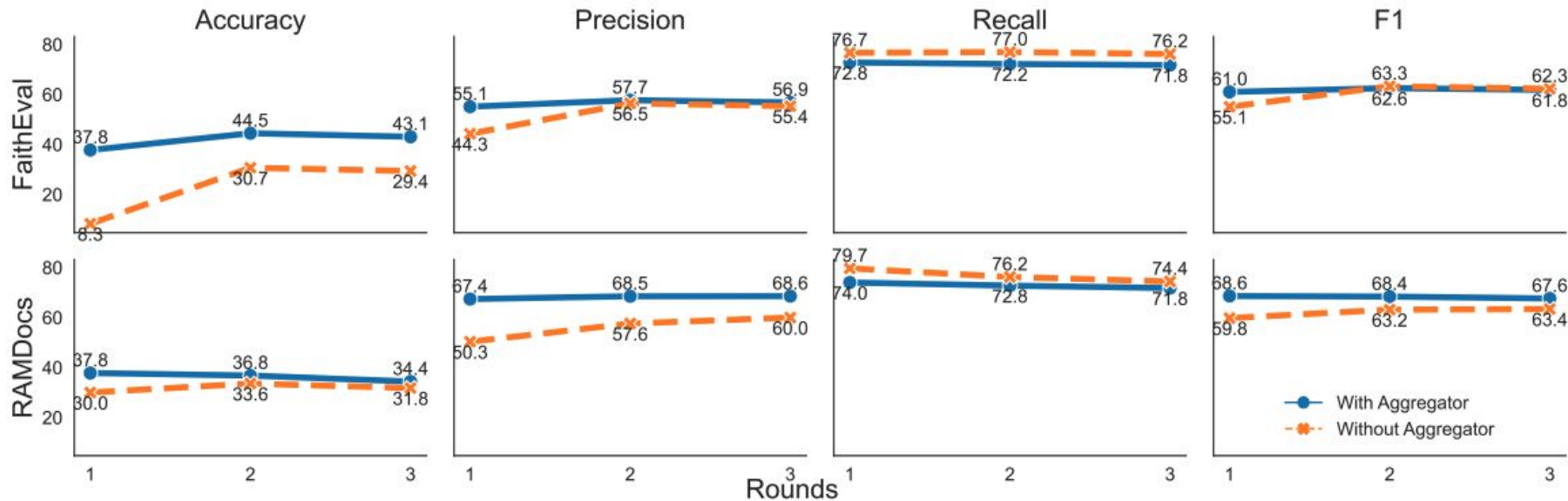
Experiments

- 3 datasets: FaithEval (misinformation & noise), AmbigDocs (ambiguity), RAMDocs
- Metrics: exact match, correct iff includes all gold answers and no incorrect or misleading answers supported only by misinformation documents.
- Compared MADAM-RAG against:
 - No RAG
 - Concatenated-Prompt
 - Single Agent with Self-reflection
 - Self-RAG
 - Speculative RAG
 - Astute RAG

Results

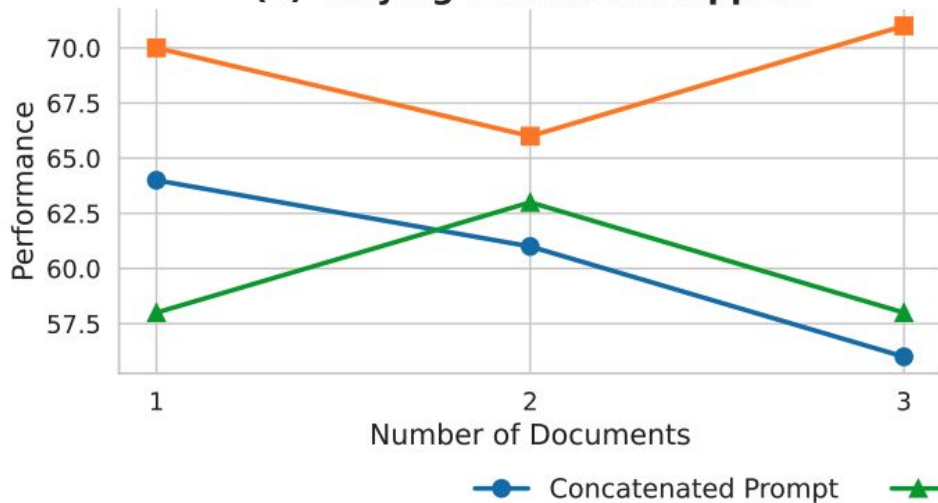
Model	Method	FaithEval	AmbigDocs	RAMDocs
SelfRAG-Llama2-13B	Self-RAG	12.40	55.00	26.80
Llama3.3-70B-Inst	No RAG	26.70	4.30	5.80
	Prompt-based	27.30	54.20	32.60
	Single Agent with Self-reflection	25.70	54.50	30.40
	Speculative RAG	41.80	44.30	30.60
	Astute RAG	37.10	46.80	31.80
	MADAM-RAG	43.10	58.20	34.40
Qwen2.5-72B-Inst	No RAG	26.40	1.80	4.20
	Prompt-based	38.50	41.20	20.60
	Single Agent with Self-reflection	39.40	29.40	21.80
	Speculative RAG	56.20	13.40	22.20
	Astute RAG	44.60	39.80	20.80
	MADAM-RAG	57.70	52.70	26.40
GPT-4o-mini	No RAG	31.00	1.00	2.50
	Prompt-based	21.00	51.50	25.00
	Single Agent with Self-reflection	36.00	44.50	22.50
	Speculative RAG	36.50	22.50	23.00
	Astute RAG	34.00	15.00	13.00
	MADAM-RAG	38.50	63.00	28.00

Ablations - Aggregator & Multiple Rounds

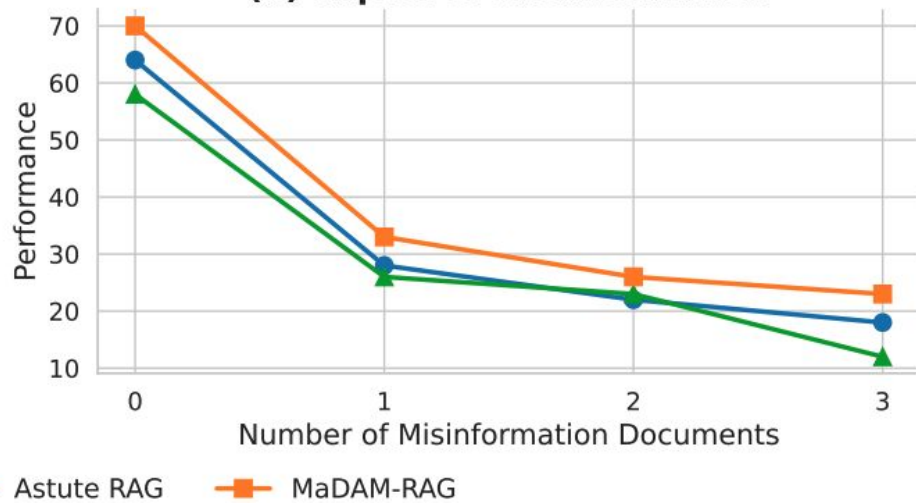


Ablations - Varying Document Support & Misinformation

(a) Varying Document Support



(b) Impact of Misinformation



Conclusion

- New dataset covers conflict due to query ambiguity, misinformation, noise all at the same time.
- New multi-agent approach to handle these conflicts through agent debate
- Point to future directions in RAG systems

GAIA

A Benchmark for General AI Assistants

Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, Thomas Scialom

World of AI benchmarks

- Trend of the search for more challenging benchmarks
- LLMs are assessed on tasks more difficult for humans
 - Many tasks difficult for humans are close to be solved
 - Human evaluation becoming less feasible
 - Model-based evaluation requires a “stronger” model, not feasible for SOTA models
- Instead AI systems can be tasked with conceptually simpler tasks that require accurate execution of complex sequence of actions
- Analogous to Proof of Work – complex problem whose solution is easy to verify

GAIA

A benchmark for General AI Assistants featuring 466 crafted questions and their answers + associated design methodology

Easy to create, simple answer easy to check, yet challenging for AI systems

Aims to avoid usual LLM evaluation pitfalls

Design Choices

- Qs are conceptually simple, rooted in real-world and challenging questions like browsing the web, using tools, handling multi-modality, etc.
- Interpretability through conceptual simplicity, easy to trace
- Non-gameable questions/robustness to memorization
- Ease of use – simple, clear prompts; concise, unambiguous answers
 - Allows for simple and fast evaluations

Example Question

GAIA Question: The attached Excel file contains the sales of menu items for a local fast-food chain. What were the total sales that the chain made from food (not including drinks)? Express your answer in USD with two decimal places.



uploaded.xlsx

GPT4 (Advanced Data Analysis mode):

Working:

```
import pandas as pd

# Load the Excel file
file_path = '/mnt/data/uploaded.xlsx'
df = pd.read_excel(file_path)
```

Working:

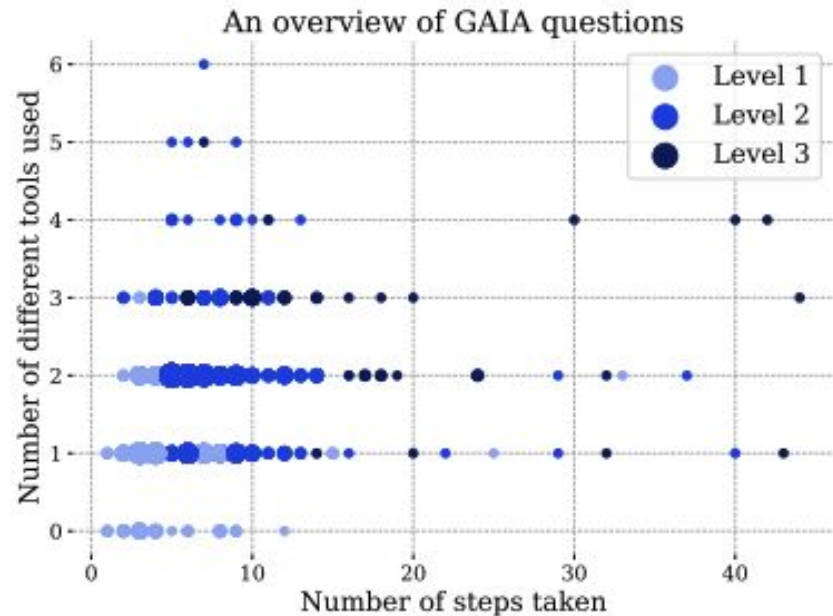
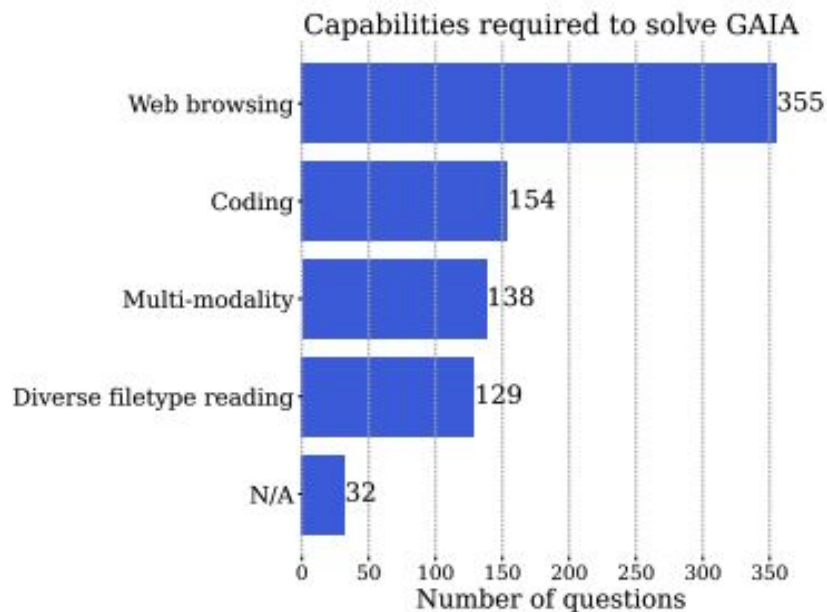
```
# Calculate the total sales from food items (not including drinks i.e., Soda)
total_food_sales = df[
    ['Burgers', 'Hot Dogs', 'Salads', 'Fries', 'Ice Cream']
].sum().sum()

# Format the result to USD, with a dollar sign and two decimal places
total_food_sales_formatted = f"${total_food_sales:,.2f}"
```

FINAL ANSWER: \$89706.00

Ground truth: 89706.00 ✓

Capabilities Coverage



Levels of Difficulty

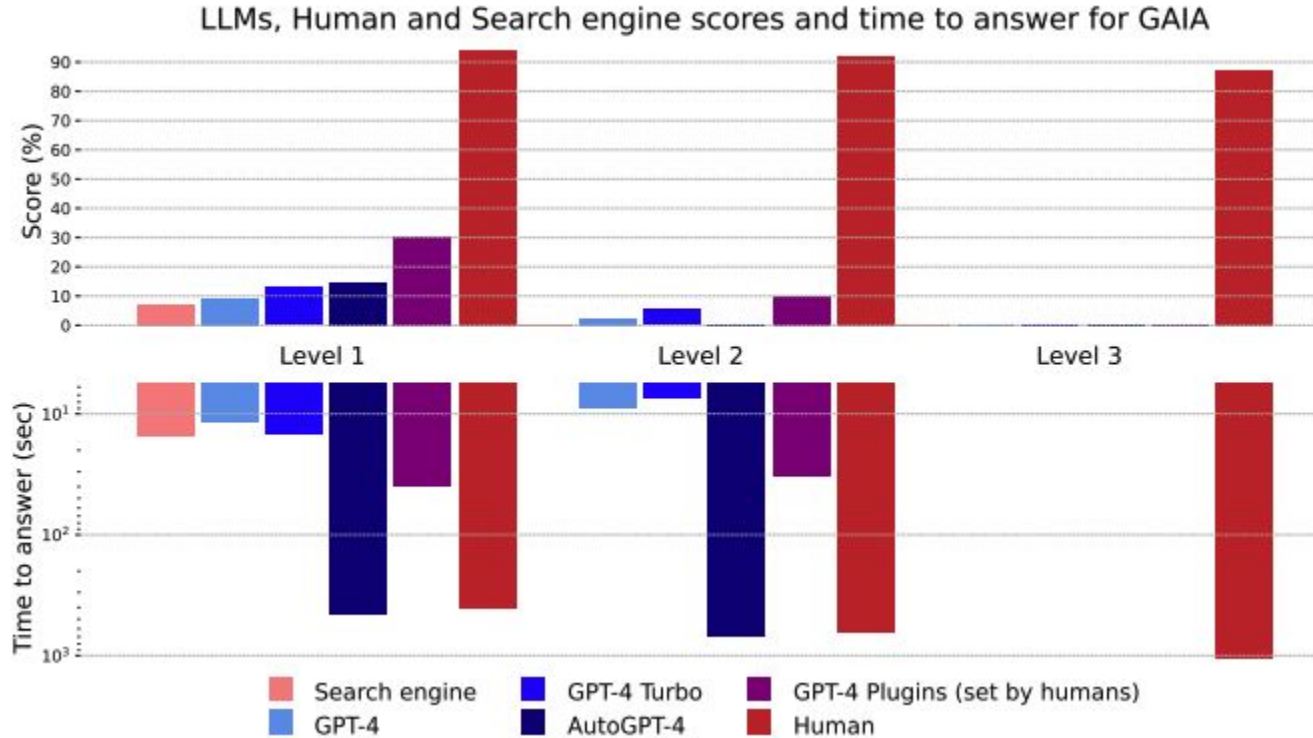
- Level 1 questions generally require no tools, or at most one tool but no more than 5 steps.
- Level 2 question generally involve more steps, roughly between 5 and 10 and combining different tools is needed.
- Level 3 are questions for a near perfect general assistant, requiring to take arbitrarily long sequences of actions, use any number of tools, and access to the world in general

Building GAIA

- Questions created by humans
 - 2 hours per question on average
- Validated by at least 2 others
- Challenges with web-based questions



LLM Results



Leaderboard

Model family	organisation	Average score (...)	Level 1 score (...)	Level 2 score (...)	Level 3 score (...)	Submission date ↑
GPT4 + AutoGPT	AutoGPT	4.98	15.05	0.63	0	2023-11-03
GPT4	GAIA authors	3.99	9.68	1.89	0	2023-11-09
GPT4	GAIA authors	6.64	9.68	6.92	0	2023-11-14
GPT3	GAIA authors	2.66	4.3	1.89	2.04	2023-11-17
GPT-4-turbo	OS-Copilot	24.25	40.86	20.13	6.12	2024-01-24
GPT-4-turbo	OS-Copilot	21.59	36.56	17.61	6.12	2024-01-30
		8.64	17.2	6.29	0	2024-02-19
		9.3	18.28	6.92	0	2024-02-20
		8.97	17.2	6.29	2.04	2024-02-20
GPT-4-Turbo		12.96	22.58	10.69	2.04	2024-02-22

<https://huggingface.co/spaces/gaia-benchmark/leaderboard>

Leaderboard

Model family	organisation	Average score ... ↕	Level 1 score (...)	Level 2 score (...)	Level 3 score (...)	Submission date
ZTE Nebula LLM, Claude Sonnet 4, Gemini 2.5 Pro	ZTE-AICloud	87.04	95.7	84.91	77.55	2025-10-13
JoinLLM, GPT-4.1, DeepSeek V3.1, Gemini-2.5-Pro	JoinAI	86.71	95.7	83.65	79.59	2025-10-09
	ch_stone	85.71	95.7	83.65	73.47	2025-10-19
GPT5, o3, Claude Sonnet 4.5, Gemini 2.5 Pro		85.05	95.7	82.39	73.47	2025-10-13
		84.72	95.7	82.39	71.43	2025-10-13
		84.72	95.7	82.39	71.43	2025-10-13
GPT5, o3, Claude Sonnet 3.7, Gemini 2.5 Pro		84.39	95.7	82.39	69.39	2025-10-10
GPT5, o3, claude sonnet 4.5, gemini 2.5 pro		84.39	95.7	82.39	69.39	2025-10-10
ZTE Nebula LLM, Claude Sonnet 4, Gemini 2.5 Pro	ZTE-AICloud	84.39	95.7	83.02	67.35	2025-09-24
GPT5, o3, Claude-Sonnet-3.7, Gemini-2.5-Pro		84.05	95.7	82.39	67.35	2025-10-10
		84.05	95.7	82.39	67.35	2025-10-10
Claude Sonnet 4, Gemini 2.5 Pro	ZTE-AICloud	84.05	95.7	83.02	65.31	2025-08-30

<https://huggingface.co/spaces/gaia-benchmark/leaderboard>

Thank You!